

Sanjeev Kumar

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1. Executive Summary

Sanjeev Kumar holds a B.Tech. in Electronic and Communications Engineering from Punjab Technical University in India. He has 15 years of industrial experience in designing, developing and implementing electronic and embedded systems for various real-world applications. He achieved in total 3 international journal publications, 2 publications with University of Cambridge, UK, 1 publication/ 2 invention patents with HRG, Huzhou China and is a member of BIS (Bureau of Indian Standard), PGD18-Industrial Automation Systems And Robotics Sectional Committee.

2. Higher Education

Period	Qualification	Subject	Organisation	Marks
08/2007-06/2011	B. Tech	Electronic & Comms Eng	KC College of Eng & IT, Punjab Tech Univ, India	75%
03/2005-06/2006	Intermediate	Physics, Chemistry, Math	Bokaro Ispat Senior Sec. School, Central Board Sec Edu, Jharkhand, India	74.6%

3. Employment

Dec 2024-Apr 2026	Embedded/Software Engineer, HRG Huzhou China
Oct 2022-Nov 2024	Technical Director, Endoenergy Systems Pvt. Ltd., Mohali, India
Jun 2022-Sep 2022	Senior Engineer, Endoenergy Systems Pvt. Ltd., Mohali, India
Feb 2020-May 2022	Junior Engineer, Endoenergy Systems Pvt. Ltd, Mohali, India
Jan 2018-Feb 2020	IoT/Embedded Developer, Allsoft Solutions Pvt. Ltd. (IBM partners), Mohali, India.
Jan 2015-Dec 2017	Embedded Developer, Jetking Chandigarh, Punjab, India
Jan 2013-Dec 2014	Senior Engineer (Electronics), A2IT Pvt. Ltd., Mohali, India
Aug 2011-Dec 2012	Embedded Apps Engineer, Vibe Tech Solutions Pvt. Ltd., Mohali, India

4. Professional Activities

Dec 2024-Apr 2026	MUSIC project: Modular Exoskeletons & User Safety Innovations in China. Wearable assistive exoskeleton R&D project: Details of professional activities and my achievements using STM32 development board Nucleo F767ZI - Set-up the electrical/ electronic lab with fully operational functionalities. - Mock-up rig: Responsible for selecting sensors, motors and wiring the mock-up rig developed to test mainly the load cells signals while donning and doffing including IMU signals and motor data using in-built encoder. The design was done by the mechanical team and it was noticed that the rig is
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	sliding down on human body and creating misalignment while movement with velcro. • Experimental rig: Responsible for wiring to perform the control test and airbags test for comfort. It was noticed that the airbag option is adding comfort but the hand soldering wires are not good options while movement as it is breaking very often. In future ready-made wires and connectors manufactures to be considered. • Full Desktop set-up: Solely responsible for connecting 2 leg sensors-motors wiring and programming to mimic the product prototype on desktop to debug and study real-time data for data collection. • Alpha product prototype development: Responsible for selecting all electrical and electronics requirements and initiated requirements to design PCB as Nucleo F767ZI size is big and not good enough to put in the product prototype.
Feb 2020-Nov 2024	Wearable assistive exoskeleton R&D: Details of professional activities and my achievements using STM32 development board Nucleo F767ZI • Developed sensor harness rig to study human knee motion and cross verify different sensors to calculate knee angle as an individual and in a team with mechanical engineers • Integrated as an individual and in a team to develop motor rig to study motor characteristics and motor control for torque test and power estimation test • Responsible for integrating electronics in Mock-up design done by mechanical engineers to make it operational with different modes • Solely responsible to integrate latest motor rig for our product with all electrical, electronics and programming to make it functional • Search, selection and procurement of components with BoM • Leading the team and technical work
Aug 2011-Feb 2020	Embedded systems training, R&D, Projects: Details of professional activities and my achievements • Trained 250+ candidates including students, teachers, professors, professionals and industry experts on embedded systems, IOT and robotics. Independently made projects list are as under using STM32, Raspberry Pi, Atmega 328P, AVR, PIC, 8051 • Alcohol detector: detects alcohol and on the buzzer • LPG & Smoke detector: helps to detect cooking gas in kitchen • Home Automation: Mobile control home lights using Bluetooth • Automatic Train barrier prototype: barrier on and off using IR • Traffic light prototype system: counters with RYG, LED's • Remote control robot: RF based remote control 4-wheel robot • Propeller display: rotating LED based name display

	• Real time clock: Display original time using DS1307 RTC module • Digital clock: HH:MM: SS clock on LCD display • Dot-matrix rotating name display: 8x8 LED matrix using shift reg • Fire alarms: on the buzzer when fire detected using fire sensor • Line follower robot: follow the black strips line 2-wheel robot • Dual way person counter: count incoming and outgoing both • DTMF robot: Keypad mobile based 4-wheel robot movements • Digital voting machine: 4 tag switch and LCD based voting with deciding winners
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5. Honours/ Awards/ Recognitions

November 2022	Memento by UPES Dehradun to organise Hackathon
December 2020	Employee of the month at Endoenergy Systems Pvt. Ltd.
August 2019	Earned certificate by IBM as Artificial Intelligence Analyst
October 2010	Represent the college as in Core Technical Committee
June 2005	Won fifth prize in Inter-school Chess competition at state level

6. Engineering Projects and Skills

1. Electrical engineering

- a. BLDC motor: Motor position, velocity and current control using STM32 via CAN bus; experimental motor torque estimation using different loads to validate motor characteristics.
- b. Motor position control data collection and analysis using serial monitor for angle, current, RPM, temperature
- c. Battery charging and BMS design: Developed BMS design for Li-ion cells for 2 series circuit for 10A. Collected data for single cell Li-ion charging and discharging at different current from 1-10A using serial monitor and plotted graph using MATLAB

2. Electronic engineering

- a. Analog circuit design using OP-AMP (LM 324/358), Transistors (BC547/548), decoders CD4017, 555 timers: created basic circuits like flip-flop, led chaser and automatic on-off circuit to test different aspects of the modules available
- b. Digital circuit design: Designed all microcontroller-based projects discussed first on software Proteus to test simulation, and then soldered the circuits to be in live mode
- c. Electronic subsystem design: Designed for exoskeleton system diagram, functional diagram, circuit diagram at sub system level and holistic level to integrate it with mechanical parts to realise it as a product

3. Embedded engineering

- a. Microcontroller development board knowhow: STM32, Raspberry Pi, Arduino, AVR, PIC, 8051: Programming using Embedded C from scratch to expert level to interface different components starting from LED to sensors, motor all possible devices known to study, interface and collecting real data
- b. IMU (ICM 20948) sensor know-how: Integrated, calibrated, and measured angles via IMU working on I2C protocols using STM32: It is being used to

measure and study knee angle during exoskeleton rig development. Also worked upon MPU6050 and MPU 9250.

- c. Load cell sensor (FX29K0/HY928-DA/913-EC) know-how: Integrated & calibrated load cell to measure force working on I2C/GPIO and to rotate motor clockwise and anticlockwise via CAN using STM32: It is being used to measure and study human intention to control motor during exoskeleton rig development
- d. Temperature sensors (NTC 10k) know-how: Integrated and calibrated multiple thermistors to get readings up to 0.1 degree accurate with single ADC channel in STM32, DHT11/22, DS18B20: It is being used to measure and study the temperature of Li ion cells inside battery testing chamber being developed to estimate the temperature at different terminals and the chamber temperature
- e. EMG sensors know-how: to measure muscle signal via ADC using STM32: It is in initial testing stage, used to measure and study muscle using conducting pads attached through skin and collecting real data while compressing and stretching

4. Software engineering

- a. Embedded C programming: STM32 Cube IDE, Keil u vision, MPLAB IDE 8.83, AVR Studio, Win AVR: used for developing software for different development boards available in the market and suitable environment supported to program it to interface different components and modules available
- b. Electrical Simulation: Proteus, to test all circuits before soldering and to save all electronics devices before integrating to check its operation on software with full functionality
- c. PCB designing software: KiCAD basics
- d. AI software implementations: IBM Watson, used to test natural language processing, chatbot and IBM Blue-mix cloud connectivity using Raspberry Pi platform
- e. Computing Software: MATLAB used it to study and implement image processing and image acquisitions toolbox. Guided three students from Shoolini University, Solan Himachal Pradesh, India to complete their thesis
 - i. Thesis1: Image de-noising using non-local algorithms.
 - ii. Thesis2: Colour compensation for vision defects.
 - iii. Thesis3: Image enhancing using texture synthesis.

7. Admin/ Finance/ Management Experience

1. Admin: Providing support for project management for technical and non-technical tasks, arranging meetings and producing meeting minutes, recruitment of engineers and interns, reviewing CV, shortlisting, interviewing. Report presentation preparation.
2. Finance: Providing support for creating supply chain and optimization of costs
3. Management: Managing engineers and interns. Project management tools implemented via Atlassian product: JIRA, Confluence and Slack.

8. Technical Projects Completed

Project title and dates	Aim and Achievements
2025-2026	• Sensor integration and calibration test using STM32: Encoder, Load cell & IMU ○ 14-bit magnetic encoder in-built in motor used to get rpm, position, current and temperature of the motor ○ Load cell: Various load cells FX29K series, HY928-DA and 913-EC are used to perform known weight

	test for calibration from 0.5 kg to 5kg for comparisons ◦ ICM 20948: used to performed the magnetic field test including known magnet, gaussmeter and itself magnet. Used to accelerometer, gyroscope and magnetometer readings and using DMP library to get raw, pitch and yaw. • BLDC motor CubeMars AK10-9 V2.0 KV60/KV100 series control using CAN protocol through STM32: Position & Current control, PID control, open loop test, step test. Performed CANalyzer test for debugging using CANalyst-II • GUI: Image and text on OLED GC9A01 using SPI and STM32 for real sensors and motor data. STM32CubeMonitor for live graph plotting for cross validation • GUI: Text and graphs on ST7789 TFT LCD 2.4"/2.0" using SPI and STM32 for real sensors. Tested LVGL library
2024	Sensor-Motor Actuator control rig for exoskeleton development • Aim is to study the shared BLDC motor control for exoskeleton development • BLDC motor Cubemars AK80-64 KV80 position and current control using CAN protocol 29-bits CAN2.0B
2023	Motor torque calibration using STM32 for exoskeleton development • It was done for power estimation and power limit test of motor • It was achieved that motor torque was greater than datasheet record
2018-2022	Ultrasonic RADAR • It was made to detect the object using ultrasonic sensor and Atmega 328P • It was achieved that ultrasonic sensor model HC-SR04 can record data accurately till 3m which is less than datasheet
	Blind man stick • Aim was to guide the blind man and realise object in the path with buzzer sound • Using IR sensor, ultrasonic sensor, buzzer and Atmega 328P it was achieved and if any obstacle in the path can be observed by sensors. Sensors are placed vertically also to analyse the tall object • 9V battery used for power
	Iron man hand gesture control robot • Aim was to move a 2-wheel robot using hand gesture for forward, backward, right and left movements • Using IMU MPU 6050, Atmega 328P, RF module, chassis, wheel and geared DC motor (100RPM) it was achieved • Li-ion of 3.7V (2) are used in series to power the robot • 9V battery is used for master circuit used for gesture using IMU and Atmega 328P
	IBM Node Red based dashboard application • Aim was to update the Raspberry Pi 3 CPU temperature on cloud using IBM Bluemix cloud • It was monitored that processor was heating very often,

	need to put DC fan for proper operation
	Voice control home automation • Aim was to control home light using google assistant through mobile phone • Node MCU ESP 8266 module used using Adafruit platform providing IFTTT if this then that node to connect application • It was achieved that it will replace distance issue and we can control home lights and devices from any country, the only requirement is the internet for hotspot connection through Wi-Fi

9. Outputs/ Publications

October 2025	Virk GS, Kumar S, He J, Peng B, Occhipinti L and Bai S. Towards commercial viability of wearable exoskeletons, ASME Letters in Translational Robotics, ALTR-25-1018, October 6, 2025
March 2024	Tang C, Yi W, Kumar S, Virk GS, and Occhipinti LG, EMG-based human motion analysis: A novel approach using towel electrodes and transfer learning, IEEE Sensors Journal, Vol 24, Issue 6, pp9115-9123, March 2024
July 2023	Tang C, Xu Zhenyu, Occhipinti E, Yi W, Xu M, Kumar S, Virk GS, Gao, S, and Occhipinti L, From brain to movement: Wearables-based motion intention prediction across human nervous system, Nano Energy, 108712, 2023

10. Outputs/ Patents

17 Nov 2025	Invention patent: Natural user interface for wearable exoskeletons. Inventor: Virk GS, Z Shen and Kumar S Owner: HRG, Huzhou China Application Number: 2025116727461 Publication Number: 2025111401675290 Application date: 14 Nov 2025 Accepted date: 17 Nov 2025
21 Jan 2026	Invention patent: Airbag component for improving the comfort of wear duration of mobility exoskeletons Inventor: Virk GS, S Zhou, Kumar S and Yuwe T Owner: HRG, Huzhou China Application Number: 2026100813993 Publication Number: 2026012101340370 Accepted date: 21 Jan 2026

11. Invitations/ Collaborations

1. May 2023, Invited for online presentation on wearable exoskeletons, Lincoln University, UK.
2. Jan 2023, DZP Technologies Ltd., UK for technical discussions on sensors and battery technologies.
3. Jan 2023, Invited by University of Cambridge, UK for collaborative project on "Wearable smart healthcare systems"
4. Jan 2021 – Nov 2022, Collaboration with UPES University to train engineering and computer science students (Undergraduate and Post-graduate) in areas of embedded systems

5. Jan 2018 – Mar 2019, Collaboration with Chandigarh University, Gharuan, Punjab to train engineering and computer science undergraduate students in areas of embedded systems, robotics and internet of things
6. Mar 2014 – May 2015, Assisted Rockwell Industries Plants Limited, Lalru Sarsini, Punjab, India in sterilizer development for the Indian Army.

12. Leisure Interests

Deeply involved in traditional and ancient Indian scriptures Vedas, Puranas, Astrology, Vastu-shastra, Numerology, colors, gemstones, metals etc.

I hereby declare that the above information is true to the best of my knowledge.

Sanjeev Kumar

Date:

Place: